Wind Forecasting with Complex ARIMA

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Abstract

Abstract text.

Keywords:

1. Introduction

Modelling and forecasting environmental variables have a broad range of applications with substantial impact on human activity. For instance, rain and ambient temperature are of crucial importance to agriculture, while solar irradiance and wind are increasingly important to energy systems. The environmental variables may be more or less difficult to model and predict. For wind, a key characteristic is that it is not a single variable, but two instead: it could be wind speed and wind direction, or its zonal and longitudinal components. Wind is highly variable in time, with seasonal effects and intrinsic nonstationarities and correlated in space. It is a by-product of many intertwined atmospheric processes, combining synoptic, mesoscale and local effects. Impact areas for wind forecasting are many, from sailing to insurance and the construction industry. Though today, it is clearly its application to power and energy systems that drives further proposal and refinement of new forecasting approaches.

In energy applications, from turbine control on a time scale of seconds to electricity market participation on a time scale of days, forecasts should only inform about the conditional expectation, but provide a full predictive distribution instead (and possibly its dependence across time, space and direction). Research on wind forecasting is interlinked with the general developments in weather and climate forecasting. However, until the early 1990s, wind was mainly seen as a by-product variable, and not a variable that would receive a dedicated focus. This has changed over the last 30-40 years, with a wealth of new approaches being proposed and benchmarked. These rely mostly on time series models, spatio-temporal models and machine learning (including all various flavours being pushed forward in the recent years). Most of these share the same limitations, i.e., wind is treated either as a scalar speed or as a pair of loosely coupled scalars (speed and direction), rather than as the single two-dimensional vector quantity it physically is. Our core proposal here is to that complex-valued random variables and processes offer a natural, and still largely under-exploited, framework for overcoming this limitation.

The earliest and still most widely used statistical descriptions of wind are univariate time-series models. Wind speed at a single site is commonly modelled (possibly after stabilising variance, or using various transformations) with autoregressive moving-average (ARMA) type processes. A recurring empirical observation is that the dynamics of wind are not homogeneous in time: different regimes alternate, owing to synoptic, mesoscale and local conditions, and the appropriate autoregressive coefficients and innovation variance ought to change accordingly.

This has motivated regime-switching formulations, most notably Markov-switching autoregressive (MSAR) models, in which a hidden Markov chain governs which of several local linear models is active at each time step (Ailliot and Monbet, 2012; Pinson and Madsen, 2012). These models allow modelling rapid changes in process dynamics, as well as changes in the variance of predictive densities. A drawback is that such models may be very sensitive at the estimation stage and difficult to use in an online setting, i.e., when data is streaming, and when one wants to adapt parameters any time new data becomes available. Additionally, all these univariate time-series models have to be proposed and estimated for each and every site individually. This comes at a high cost, while also over-looking potential spatio-temporal effects (which are widely know for wind fields).

Spatio-temporal models were developed to close this gap by treating collections of sites jointly. Vector extensions of the autoregressive and Markov-switching frameworks allow the state, and even the switching regime, to be shared across a network of stations or turbines, so that spatial dependence and space-time propagation of weather systems are represented explicitly (Hering et al., 2015). A further refinement, motivated by the recognition that wind speed and wind direction should not be modelled independently, replaces the scalar wind field by a bivariate one and introduces cross-covariance, or coregionalization, structures that couple the two components in space and time; recent work along these lines has produced substantial gains in the reliability of probabilistic wind vector forecasts (Lenzi and Genton, 2020). These bivariate spatio-temporal models make an important conceptual step, acknowledging that speed and direction are two faces of a single vector process, but they typically still require bespoke, high-dimensional cross-covariance matrices to link the two components, chosen among a limited catalogue of valid constructions, and this construction becomes markedly more involved as the number of monitored sites grows.

Alongside these classical formulations, the last decade has seen a rapid expansion of machine learning and, in particular, deep learning approaches to wind modelling and forecasting. Recurrent architectures such as long short-term memory (LSTM) networks, convolutional networks operating over turbine or grid-point layouts, and, more recently, graph neural networks that exploit the topology of wind farm clusters, have all been reported to improve forecast accuracy relative to purely statistical benchmarks, particularly at short lead times and in the presence of complex terrain or wake effects. Hanifi et al. (2020) provide a comprehensive recent review of this literature alongside the physical and statistical approaches it complements. These methods place very few a priori restrictions on the functional form of the temporal and spatial dependence, and can absorb heterogeneous covariates with little manual feature engineering. Their cost, however, is a general lack of interpretability, an appetite for large training data, and, importantly here, no inherent notion that speed and direction are components of one physical vector: like most time series and spatio-temporal models, they are almost always applied either to wind speed alone or to speed and direction as two separately engineered inputs.

This recurring separation of speed and direction, or, equivalently, of the two orthogonal horizontal wind components, comes with major drawbacks. It necessarily discards information regarding the dependence between the two. These two components are generally correlated and jointly non-Gaussian. There may also be a marked anisotropy induced by prevailing wind directions and local terrain effects. Directional data of this kind are traditionally handled with the tools of circular statistics, or, in the bivariate spatio-temporal setting above, with explicitly constructed cross-covariance functions. Complex-valued random variables and processes offer a more natural and considerably more economical alternative. This is since the wind vector can be

expressed as a single object (Z, a complex number) for which the real and imaginary parts are the zonal and meridional components, respectively. A whole toolbox from complex-valued statistical signal processing, developed originally for communications and array processing, is then available to model such complex-number based processes. Complex Gaussian and elliptical distributions (Goodman, 1963), complex-valued autoregressive and state-space models, and, crucially, second-order noncircularity, describing how the variance of Z differs across directions rather than being rotationally symmetric, provide a compact way of encoding exactly the anisotropy that drives prevailing-direction effects in wind. Ordinary complex-valued linear estimation is blind to this noncircular structure; the widely linear, or augmented, framework of Picinbono and Chevalier (1995) extends linear prediction to make full use of both the covariance and the pseudo-covariance of Z, and has already been shown, in an early application to wind profile prediction using augmented complex least-mean-square algorithms, to outperform component-wise real-valued prediction of the same data (Mandic et al., 2009).

Citing papers by Gneiting, Grimit, etc.

2. Literature Review

Papers to read:

- Goh et al. (2006): `https://www.sciencedirect.com/science/article/abs/pii/S096014810500218`
- Mandic et al. (2007): `https://www.commsp.ee.ic.ac.uk/~mandic/research/WL_Complex_Stuff/MLSP_2007_Complex_for_Real.pdf`
- Mandic et al. (2009): `https://www.commsp.ee.ic.ac.uk/~mandic/research/wind/Danilo_Renewable_Energy_ACLMS_2009.pdf`

For an overview of potential data, this one could be good (Effenberger and Ludwig, 2022).

This link also provides a good overview of datasets (which are more about wind power than wind): `https://github.com/sltzgs/OpenWindSCADA` .

I think the best could be to extract a dataset from ECMWF

3. Methodology

We use complex ARIMA developed by Svetunkov and Svetunkov (2024).

Note: If $x + iy$ follows the complex normal distribution (similar to the bivariate normal), the magnitude $R = \sqrt{x^2 + y^2}$ follows the Rayleigh distribution, while the angle $\phi = \arctan\left(\frac{y}{x}\right)$ follows the uniform one. We want to model and forecast the wind speed and direction. To do this, we use Wind Turbine Scada Dataset (https://www.kaggle.com/datasets/berkerisen/wind-turbine-scada-dataset).

This database, like many similar ones, provides data on wind speed v and direction θ. Wind direction θ is given in degrees, ranging from 0 to 360 degrees. Using this data directly for modelling can yield meaningless results.

The primary reason for this is that wind directions of 0 degrees and 360 degrees represent the same direction; similarly, directions of 1 degree and 359 degrees are virtually identical, yet their numerical representation in the domain of real numbers reveals a vast difference between the values.

The second reason is that the wind direction is measured on an interval scale limited to the range of 0 to 360 degrees. Values such as -16 or 456 degrees have no meaning. The aforementioned difficulties can be avoided by describing the wind direction using a unit vector in the complex plane.

$$\phi = cos\theta + isin\theta \tag{1}$$

Taking this into account, the wind characteristics at time t can be described using a complex variable:

$$Z(t) = v(t)e^{i\phi} \tag{2}$$

Wind speed and direction at the current time t are strongly determined by the wind speed and direction at previous times $t-1$, $t-2$, $t-3$,..., $t-p$. This relationship can be represented as an autoregression of order n:

$$Z_t = a_0 + \sum_{j=1}^{p} a_j Z_{t-j} \tag{3}$$

Here, the constant term a_0 and the proportionality coefficients a_j are complex:

$$a_0 = a_{i0} + ia_{r0} \tag{4}$$

$$a_j = a_{ij} + ia_{rj} \tag{5}$$

Instead of a simple complex autoregression, an ARIMA model should be used. .....

the logarithms of the complex variable $Re^{i\phi}$, where R is the speed of wind and ϕ is the angle.

4. Evaluation

Potential datasets:

1. `https://github.com/dominodatalab/reference-project-wind-turbine/blob/release-1.0.0/README.md`
2. `https://www.kaggle.com/competitions/predict-the-wind-speed-at-a-wind-turbine/data`

5. Conclusions

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